

NotifiED

Software Test Case Document

NotifiED Mobile Application

This document defines the test cases derived from the NotifiED Use Case Full Description. Each test case traces to its source use case and covers either the Main Flow, an Alternate Flow, or an Exception Flow. As the project is currently in the design/development stage, all test cases are documented with Status “Not Executed” and will be run once a testable build is available.

Document Overview

Modules covered:

- Manage User Account (UC-01 – UC-05)
- View Calendar (UC-06 – UC-08)
- View Email (UC-09 – UC-10)
- Browse Organization (UC-11 – UC-14)
- Access Dashboard (UC-15 – UC-17)
- Manage Events (UC-18 – UC-22)
- AI Chatbot (UC-23 – UC-24)

Field guide:

- **Test Data** — concrete sample values used to execute the test (accounts, IDs, inputs).
- **Expected Result** — the outcome the system should produce if it behaves correctly.
- **Priority** — relative importance of the test case (High / Medium / Low).
- **Status** — execution state of the test case (Not Executed / Passed / Failed).

Test Case Summary:

Module	Use Cases Covered	Test Case Count
Manage User Account	UC-01 to UC-05	16
View Calendar	UC-06 to UC-08	8
View Email	UC-09 to UC-10	5
Browse Organization	UC-11 to UC-14	9
Access Dashboard	UC-15 to UC-17	12
Manage Events	UC-18 to UC-22	20
AI Chatbot	UC-23 to UC-24	7
Total		77

Manage User Account

UC-01: Log In

Test Case ID	TC-01-01
Use Case	UC-01: Log In
Test Scenario	Log In — Main Flow (Positive Scenario): Student logs in successfully with valid institutional Microsoft account
Preconditions	Student has a valid institutional account; app has active internet connection.
Test Steps	1. Open the NotifiED mobile app. 2. Tap Sign In. 3. Enter valid institutional Microsoft credentials when prompted (UC-02). 4. Wait for authentication to complete.
Test Data	Valid institutional Microsoft credentials
Expected Result	Student is authenticated, an active session is established, and the main NotifiED Dashboard loads.
Priority	High
Status	Not Executed

Test Case ID	TC-01-02
Use Case	UC-01: Log In
Test Scenario	Log In — A1: Returning student with a cached session skip sign-in
Preconditions	Student previously signed in and is within the active session window.
Test Steps	1. Open the NotifiED app within the cached session window.
Test Data	N/A
Expected Result	System detects the valid cached session token, skips the sign-in prompt, and loads the dashboard directly.
Priority	Medium
Status	Not Executed

Test Case ID	TC-01-03
Use Case	UC-01: Log In
Test Scenario	Log In — E1: Student cancels or dismisses authentication
Preconditions	Student has selected Sign In and the Microsoft sign-in prompt is displayed.
Test Steps	1. Tap Sign In. 2. On the Microsoft sign-in prompt, tap Cancel or close the prompt.
Test Data	N/A

Expected Result	System terminates authentication, displays a cancellation message, and remains on the sign-in landing screen. Use case ends in failure.
Priority	Medium
Status	Not Executed

UC-02: Authenticates User

Test Case ID	TC-02-01
Use Case	UC-02: Authenticates User
Test Scenario	Authenticates User — Main Flow (Positive Scenario): Microsoft Identity Platform verifies valid credentials and issues a token
Preconditions	Microsoft Identity Platform is available; student has valid institutional credentials.
Test Steps	1. System redirects the authentication request to the Microsoft Identity Platform. 2. Student enters valid institutional credentials. 3. Platform validates the credentials.
Test Data	Valid institutional username/password
Expected Result	Microsoft Identity Platform issues an authorization token to NotifiED; student identity is verified.
Priority	High
Status	Not Executed

Test Case ID	TC-02-02
Use Case	UC-02: Authenticates User
Test Scenario	Authenticates User — E1: Student enters invalid credentials then retries successfully
Preconditions	Microsoft Identity Platform is available.
Test Steps	1. Student enters invalid institutional credentials. 2. Observe the platform's response. 3. Student re-enters valid credentials.
Test Data	Invalid credentials, then valid credentials
Expected Result	Platform rejects the first attempt, and system displays an error message; on retry with valid credentials, use case resumes and a token is issued.
Priority	High
Status	Not Executed

Test Case ID	TC-02-03
Use Case	UC-02: Authenticates User
Test Scenario	Authenticates User — E2: Authentication fails because the identity service is unreachable

Preconditions	Microsoft Identity Platform is unavailable or unreachable (simulate outage/no network).
Test Steps	1. Student attempts to sign in. 2. System attempts to connect to the Microsoft Identity Platform.
Test Data	N/A (Service outage simulated)
Expected Result	System fails to connect, displays a service error message, and prompts the student to try again later. No token is issued.
Priority	Medium
Status	Not Executed

UC-03: Edit Profile

Test Case ID	TC-03-01
Use Case	UC-03: Edit Profile
Test Scenario	Edit Profile — Main Flow (Positive Scenario): Student successfully creates/updates their profile
Preconditions	Student is authenticated; app has internet connection; NotifiED Web Team backend is available.
Test Steps	1. Navigate to Profile section. 2. Allow system to load existing profile (UC-04). 3. Enter/update name, student ID, year level, section, and program. 4. Submit the profile. 5. Observe validation and save confirmation.
Test Data	Valid profile details (name, student ID, year level, section, program)
Expected Result	System validates the data, sends it to the NotifiED Web Team, saves it, and displays the updated profile with confirmation.
Priority	High
Status	Not Executed

Test Case ID	TC-03-02
Use Case	UC-03: Edit Profile
Test Scenario	Edit Profile — A1: Student submits incomplete/invalid profile information
Preconditions	Student is on the Profile edit screen.
Test Steps	1. Enter incomplete or invalid values in one or more required fields. 2. Submit the profile. 3. Correct the flagged fields and resubmit them.
Test Data	Missing student ID, invalid year level value
Expected Result	System detects invalid fields, displays error indicators, and prompts correction; after correction, submission proceeds and profile are validated again.
Priority	High

Status	Not Executed
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Test Case ID	TC-03-03
Use Case	UC-03: Edit Profile
Test Scenario	Edit Profile — A2: Student cancels profile editing before submission
Preconditions	Student is on the Profile edit screen with unsaved changes and has not yet submitted.
Test Steps	1. Edit one or more profile fields. 2. Select Cancel or navigate away before submitting.
Test Data	N/A
Expected Result	System discards unsaved changes and restores the previous screen; no data is sent to the backend.
Priority	Medium
Status	Not Executed

Test Case ID	TC-03-04
Use Case	UC-03: Edit Profile
Test Scenario	Edit Profile — E1: Backend is unavailable when submitting profile changes
Preconditions	Student has submitted valid profile information; NotifiED Web Team backend is simulated as unavailable.
Test Steps	1. Complete and submit the profile form. 2. Observe system behavior when the backend request fails or times out.
Test Data	Valid profile details; backend outage simulated
Expected Result	System displays an error message informing the student about the update could not be saved; existing profile information remains unchanged.
Priority	High
Status	Not Executed

UC-04: Retrieves Profile Data

Test Case ID	TC-04-01
Use Case	UC-04: Retrieves Profile Data
Test Scenario	Retrieves Profile Data — Main Flow (Positive Scenario): Existing profile data is retrieved and displayed
Preconditions	Student is authenticated; NotifiED Web Team backend is available; a profile already exists.
Test Steps	1. Navigate to Profile section (triggers UC-04). 2. Observe the request sent to the NotifiED Web Team.

Test Data	Existing student profile record
Expected Result	System displays the retrieved profile pre-filled in the edit form.
Priority	High
Status	Not Executed

Test Case ID	TC-04-02
Use Case	UC-04: Retrieves Profile Data
Test Scenario	Retrieves Profile Data — A1: No existing profile is found for the student
Preconditions	Student is authenticated and has never created a profile.
Test Steps	1. Navigate to Profile section for a new student account.
Test Data	New student account, no profile record
Expected Result	NotifiED Web Team finds no profile record; system displays a blank Profile Setup form.
Priority	Medium
Status	Not Executed

Test Case ID	TC-04-03
Use Case	UC-04: Retrieves Profile Data
Test Scenario	Retrieves Profile Data — E1: Profile retrieval fails due to backend unavailability
Preconditions	Student is authenticated; NotifiED Web Team backend is simulated as unavailable.
Test Steps	1. Navigate to Profile section. 2. Observe request timeout/failure.
Test Data	N/A (Backend outage simulated)
Expected Result	System displays an error message and prevents editing until the request is retried.
Priority	Medium
Status	Not Executed

UC-05: Log Out

Test Case ID	TC-05-01
Use Case	UC-05: Log Out
Test Scenario	Log Out — Main Flow (Positive Scenario): Student logs out successfully
Preconditions	Student has an active session.
Test Steps	1. Select Log Out from Manage User Account section. 2. Confirm the log out prompt.

Test Data	N/A
Expected Result	System clears the local session and authorization token and returns the student to the sign-in screen.
Priority	High
Status	Not Executed

Test Case ID	TC-05-02
Use Case	UC-05: Log Out
Test Scenario	Log Out — A1: Student cancels the log out action
Preconditions	Student has an active session and has selected Log Out.
Test Steps	1. Select Log Out. 2. On the confirmation prompt, select Cancel.
Test Data	N/A
Expected Result	System dismisses the prompt; the session remains active and student stays on the current screen.
Priority	Medium
Status	Not Executed

Test Case ID	TC-05-03
Use Case	UC-05: Log Out
Test Scenario	Log Out — E1: Session token fails to clear due to device storage error
Preconditions	Student has confirmed log out; device storage error is simulated.
Test Steps	1. Select Log Out and confirm. 2. Simulate a failure while clearing the local session token.
Test Data	Simulated device storage error
Expected Result	System retries automatically and forces the sign-in screen regardless, preventing a stale session from remaining active.
Priority	Medium
Status	Not Executed

View Calendar

UC-06: View Calendar

Test Case ID	TC-06-01
Use Case	UC-06: View Calendar

Test Scenario	View Calendar — Main Flow (Positive Scenario): Student views consolidated personal schedule and campus events
Preconditions	Student is authenticated; calendar information is authorized and available.
Test Steps	1. Select Calendar. 2. Allow system to retrieve personal schedule (UC-07) and campus events (UC-08). 3. Browse calendar entries by date.
Test Data	Authorized calendar access
Expected Result	System integrates and displays personal schedule and campus events chronologically in a single calendar view.
Priority	High
Status	Not Executed

Test Case ID	TC-06-02
Use Case	UC-06: View Calendar
Test Scenario	View Calendar — A1: Calendar loads with one data source unavailable
Preconditions	Student is authenticated; one of the two data sources (personal schedule or campus events) is simulated as failing.
Test Steps	1. Select Calendar. 2. Simulate failure of one supporting source while the other responds normally.
Test Data	One source (e.g., Microsoft Identity Platform) simulated offline
Expected Result	System renders the successfully retrieved data and displays a warning banner indicating partial data was loaded.
Priority	Medium
Status	Not Executed

Test Case ID	TC-06-03
Use Case	UC-06: View Calendar
Test Scenario	View Calendar — E1: Calendar fails to load when both data sources are unavailable
Preconditions	Student is authenticated; both data sources are simulated as failing.
Test Steps	1. Select Calendar. 2. Simulate failure of both the Microsoft Identity Platform and NotifiED Web Team requests.
Test Data	Both sources simulated offline
Expected Result	System displays a calendar error screen with a Retry option.
Priority	High
Status	Not Executed

UC-07: Retrieve Calendar

Test Case ID	TC-07-01
Use Case	UC-07: Retrieve Calendar
Test Scenario	Retrieve Calendar — Main Flow (Positive Scenario): Personal schedule is retrieved successfully from Microsoft Identity Platform
Preconditions	Student is authenticated; calendar access is authorized.
Test Steps	1. Trigger a calendar request (via View Calendar). 2. System sends the request with the student's access token. 3. Platform verifies the token and returns schedule entries.
Test Data	Valid access token
Expected Result	Personal schedule data is returned and passed back to View Calendar for display.
Priority	High
Status	Not Executed

Test Case ID	TC-07-02
Use Case	UC-07: Retrieve Calendar
Test Scenario	Retrieve Calendar — A1: Expired/invalid token requires re-authorization
Preconditions	Student's access tokens are expired or invalid (simulate expired token).
Test Steps	1. Trigger a calendar request with an expired token. 2. Observe token verification results. 3. Complete re-authentication when prompted.
Test Data	Expired/invalid access token
Expected Result	System prompts the student to re-authorize their Microsoft account; after re-authentication, the calendar request resumes and succeeds.
Priority	Medium
Status	Not Executed

Test Case ID	TC-07-03
Use Case	UC-07: Retrieve Calendar
Test Scenario	Retrieve Calendar — E1: Retrieval fails when the Microsoft Identity Platform times out
Preconditions	Student is authenticated; Microsoft Identity Platform is simulated as timing out.
Test Steps	1. Trigger a calendar request. 2. Simulate a timeout from the Microsoft Identity Platform.
Test Data	Simulated platform timeout
Expected Result	System reports the failure to View Calendar, so a partial data warning can be shown.

Priority	Medium
Status	Not Executed

UC-08: Retrieve Upcoming Campus Event

Test Case ID	TC-08-01
Use Case	UC-08: Retrieve Upcoming Campus Event
Test Scenario	Retrieve Upcoming Campus Event — Main Flow (Positive Scenario): Upcoming approved campus events are retrieved successfully
Preconditions	Student is authenticated.
Test Steps	1. Trigger a calendar request (via View Calendar). 2. System requests upcoming approved events from the NotifiED Web Team for the requested date range.
Test Data	Valid date range with approved events
Expected Result	NotifiED Web Team returns the event list, which is passed back to View Calendar for display.
Priority	High
Status	Not Executed

Test Case ID	TC-08-02
Use Case	UC-08: Retrieve Upcoming Campus Event
Test Scenario	Retrieve Upcoming Campus Event — E1: Retrieval fails when the NotifiED Web Team is unreachable
Preconditions	Student is authenticated; NotifiED Web Team backend is simulated as failing/timing out.
Test Steps	1. Trigger a calendar request. 2. Simulate a failure/timeout from the NotifiED Web Team.
Test Data	Simulated backend outage
Expected Result	System reports the failure to View Calendar so a partial-data warning can be shown.
Priority	Medium
Status	Not Executed

View Email

UC-09: View Email

Test Case ID	TC-09-01
Use Case	UC-09: View Email
Test Scenario	View Email — Main Flow (Positive Scenario): Student opens institutional email successfully

Preconditions	Student is authenticated in NotifiED; institutional email service is available; student has access to their institutional email account.
Test Steps	1. Student clicks the Email button. 2. Select Open Email. 3. Allow system to redirect via UC-10.
Test Data	N/A
Expected Result	Student is redirected to and can access their institutional email service through Outlook.
Priority	High
Status	Not Executed

Test Case ID	TC-09-02
Use Case	UC-09: View Email
Test Scenario	View Email — E1: Network error prevents opening email
Preconditions	Student is authenticated; network connection is simulated as unavailable.
Test Steps	1. Student clicks the Email button. 2. Select Open Email while offline/connection is failing.
Test Data	Simulated network outage
Expected Result	System displays an offline error message with a Retry button; student remains in NotifiED.
Priority	Medium
Status	Not Executed

UC-10: Redirects to Outlook

Test Case ID	TC-10-01
Use Case	UC-10: Redirects to Outlook
Test Scenario	Redirects to Outlook — Main Flow (Positive Scenario): Student is redirected to institutional email successfully
Preconditions	Student is authenticated; institutional email service is available.
Test Steps	1. Trigger the redirect from View Email (UC-09). 2. System requests the redirect target using the authenticated session. 3. System opens the institutional email app.
Test Data	N/A
Expected Result	The institutional email app opens using the authenticated account.
Priority	High
Status	Not Executed

Test Case ID	TC-10-02
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Use Case	UC-10: Redirects to Outlook
Test Scenario	Redirects to Outlook — A1: Email service requires a separate sign-in
Preconditions	Institutional email service requires its own sign-in step.
Test Steps	1. Trigger the redirect from View Email. 2. Observe redirect to the email service's sign-in page. 3. Authenticate using institutional credentials.
Test Data	Valid institutional credentials
Expected Result	Student is redirected to the email sign-in page and, after authenticating, is granted access to the inbox.
Priority	Medium
Status	Not Executed

Test Case ID	TC-10-03
Use Case	UC-10: Redirects to Outlook
Test Scenario	Redirects to Outlook — E1: Redirect fails to open the email app.
Preconditions	Institutional email service is simulated as unreachable or the app fails to open.
Test Steps	1. Trigger the redirect from View Email. 2. Simulate failure to open the email app.
Test Data	Simulated redirect failure
Expected Result	System displays an error message with a Retry option; student remains in Notified.
Priority	Medium
Status	Not Executed

Browse Organization

UC-11: Browse Organization

Test Case ID	TC-11-01
Use Case	UC-11: Browse Organization
Test Scenario	Browse Organization — Main Flow (Positive Scenario): Student browses the organization directory successfully
Preconditions	Student is authenticated; organization information is available through the backend.
Test Steps	1. Open Organization Directory. 2. Allow system to load directory data (UC-14). 3. Browse, filter (UC-12), or view organization details (UC-13) as needed.
Test Data	N/A
Expected Result	Student successfully browses organization profiles and their associated activities.
Priority	High

Status	Not Executed
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Test Case ID	TC-11-02
Use Case	UC-11: Browse Organization
Test Scenario	Browse Organization — A1: Search returns zero matching organizations
Preconditions	Organization directory is loaded.
Test Steps	1. Enter a search query that matches no organization records. 2. Clear the search query.
Test Data	Search term with no matches, e.g., ' Nonexistent Organization '
Expected Result	System displays 'No organizations found matching your search.'; after clearing the query, the full directory is shown again.
Priority	Medium
Status	Not Executed

Test Case ID	TC-11-03
Use Case	UC-11: Browse Organization
Test Scenario	Browse Organization — E1: Organization directory fails to load
Preconditions	Student is authenticated; backend service is simulated as returning an error.
Test Steps	1. Open Organization Directory while the backend is simulated as failing.
Test Data	Simulated backend error
Expected Result	System displays an error notification stating the directory is unavailable.
Priority	Medium
Status	Not Executed

UC-12: Filter Organization

Test Case ID	TC-12-01
Use Case	UC-12: Filter Organization
Test Scenario	Filter Organization — Main Flow (Positive Scenario): Student filters organizations by category or keyword
Preconditions	Organization directory data is loaded.
Test Steps	1. Enter a search keyword or select a category filter. 2. Observe the filtered results.
Test Data	Valid category, e.g., 'Academic Organizations'

Expected Result	System applies the filter criteria (retrieving a fresh dataset via UC-14 if needed) and displays the filtered results.
Priority	High
Status	Not Executed

Test Case ID	TC-12-02
Use Case	UC-12: Filter Organization
Test Scenario	Filter Organization — A1: Filter criteria match no organizations
Preconditions	Organization directory data is loaded.
Test Steps	1. Enter a filter/category that matches no organizations.
Test Data	Filter with zero matches
Expected Result	System displays 'No organizations found matching your search.'
Priority	Medium
Status	Not Executed

UC-13: View Organization Details

Test Case ID	TC-13-01
Use Case	UC-13: View Organization Details
Test Scenario	View Organization Details — Main Flow (Positive Scenario): Student views full organization profile
Preconditions	The organization exists in the directory.
Test Steps	1. Select an organization from the directory. 2. Allow system to fetch full profile (UC-14) if not cached. 3. Review name, logo, description, category, and associated events.
Test Data	Existing organization record
Expected Result	Organization profile details are displayed, and the student can navigate to an associated event's details.
Priority	High
Status	Not Executed

Test Case ID	TC-13-02
Use Case	UC-13: View Organization Details
Test Scenario	View Organization Details — E1: Selected organization is no longer available
Preconditions	Organization record has been removed or is missing (simulate deleted org).

Test Steps	1. Select an organization that has been removed from the directory.
Test Data	Removed/deleted organization ID
Expected Result	System displays a 'This organization is no longer available' message.
Priority	Medium
Status	Not Executed

UC-14: Retrieve Organizations

Test Case ID	TC-14-01
Use Case	UC-14: Retrieve Organizations
Test Scenario	Retrieve Organizations — Main Flow (Positive Scenario): Organization data is retrieved successfully
Preconditions	Student is authenticated.
Test Steps	1. Trigger a request for organization directory data (via UC-12 or UC-13). 2. System sends the request to the NotifiED Web Team.
Test Data	N/A
Expected Result	NotifiED Web Team returns the organization list/profile, which is passed back to the requesting use case.
Priority	High
Status	Not Executed

Test Case ID	TC-14-02
Use Case	UC-14: Retrieve Organizations
Test Scenario	Retrieve Organizations — E1: Organization directory data fails to retrieve
Preconditions	Student is authenticated; backend service is simulated as returning an error.
Test Steps	1. Trigger a request for organization data while the backend is simulated as failing.
Test Data	Simulated backend error
Expected Result	System displays a notification stating the directory is unavailable.
Priority	Medium
Status	Not Executed

Access Dashboard

UC-15: Access Dashboard

Test Case ID	TC-15-01
Use Case	UC-15: Access Dashboard
Test Scenario	Access Dashboard — Main Flow (Positive Scenario): Student views the dashboard with live data
Preconditions	Student is authenticated.
Test Steps	1. Navigate to the Dashboard. 2. Allow system to load notification preview (UC-16) and active data (announcements, events, schedules). 3. Review dashboard sections.
Test Data	N/A
Expected Result	System formats and renders dashboard sections; student views summary feeds and can select items for detailed views.
Priority	High
Status	Not Executed

Test Case ID	TC-15-02
Use Case	UC-15: Access Dashboard
Test Scenario	Access Dashboard — A1: Dashboard falls back to cached data when live retrieval fails
Preconditions	Student is authenticated; live backend retrieval is simulated as failing/timing out; local cached dashboard data exists.
Test Steps	1. Navigate to the Dashboard while the backend is simulated as failing. 2. Observe fallback behavior.
Test Data	Simulated backend timeout with cached data present
Expected Result	System displays cached dashboard items with a warning banner indicating offline/cached mode.
Priority	Medium
Status	Not Executed

Test Case ID	TC-15-03
Use Case	UC-15: Access Dashboard
Test Scenario	Access Dashboard — A2: Dashboard shows empty state when no announcements or events exist
Preconditions	Student is authenticated; backend is simulated to return an empty dataset.
Test Steps	1. Navigate to the Dashboard when no announcements/events are available.

Test Data	Empty dataset from backend
Expected Result	System renders empty-state placeholders encouraging the student to check back later.
Priority	Low
Status	Not Executed

Test Case ID	TC-15-04
Use Case	UC-15: Access Dashboard
Test Scenario	Access Dashboard — A3: Student manually refreshes the dashboard via pull-to-refresh
Preconditions	Student is on the Dashboard.
Test Steps	1. Perform a pull-to-refresh gesture on the Dashboard.
Test Data	N/A
Expected Result	System sends a fresh data request to the backend and updates the displayed feed.
Priority	Medium
Status	Not Executed

Test Case ID	TC-15-05
Use Case	UC-15: Access Dashboard
Test Scenario	Access Dashboard — E1: Dashboard fails with no cached data and no network
Preconditions	Student is authenticated; connection is simulated as failing and no local cached data exists.
Test Steps	1. Navigate to the Dashboard with no network connection and no cached data available.
Test Data	Simulated total network failure, no cache
Expected Result	System displays a full-screen network error with a Retry option.
Priority	High
Status	Not Executed

UC-16: View Notification

Test Case ID	TC-16-01
Use Case	UC-16: View Notification
Test Scenario	View Notification — Main Flow (Positive Scenario): Student opens a push notification and is routed to the linked item
Preconditions	Student is authenticated and notifications are enabled.

Test Steps	1. Trigger a push notification from the Notified Web Team. 2. Receive the notification on the mobile device. 3. Tap/select the notification. 4. Allow system to retrieve related data (UC-17).
Test Data	Valid, active notification linked to an announcement/event
Expected Result	System routes the student directly to the relevant announcement or event detail page.
Priority	High
Status	Not Executed

Test Case ID	TC-16-02
Use Case	UC-16: View Notification
Test Scenario	View Notification — A1: Notifications are processed in-app when push delivery is disabled
Preconditions	Push delivery is suppressed by system permission settings.
Test Steps	1. Disable push notification permissions. 2. Open the application manually. 3. Review the in-app notification center.
Test Data	Push notifications disabled at OS level
Expected Result	System highlights unread alerts in the in-app notification center.
Priority	Medium
Status	Not Executed

Test Case ID	TC-16-03
Use Case	UC-16: View Notification
Test Scenario	View Notification — E1: Notification links to content that was deleted or expired
Preconditions	Student receives a push notification whose linked item has since been deleted.
Test Steps	1. Tap a notification whose linked announcement/event was deleted.
Test Data	Notification linked to a deleted item
Expected Result	System displays 'Linked content no longer available' and redirects to the main Dashboard.
Priority	Medium
Status	Not Executed

Test Case ID	TC-16-04
Use Case	UC-16: View Notification
Test Scenario	View Notification — E2: Student opens a notification while offline

Preconditions	Student receives a notification while the device has no internet access.
Test Steps	1. With internet access disabled, tap/open the notification.
Test Data	Simulated offline device
Expected Result	System detects offline status and displays a connection error screen.
Priority	Medium
Status	Not Executed

UC-17: Retrieve Data

Test Case ID	TC-17-01
Use Case	UC-17: Retrieve Data
Test Scenario	Retrieve Data — Main Flow (Positive Scenario): Requested campus data is retrieved successfully
Preconditions	Student is authenticated.
Test Steps	1. Calling use case (e.g., View Notification or Browse Events) sends a data request specifying the type of information needed. 2. System forwards the request to the NotifiED Web Team.
Test Data	Valid data type (announcement, notification, or event summary)
Expected Result	NotifiED Web Team retrieves and returns the requested data, which the system passes back to the calling use case.
Priority	High
Status	Not Executed

Test Case ID	TC-17-02
Use Case	UC-17: Retrieve Data
Test Scenario	Retrieve Data — A1: Live retrieval fails and falls back to local cached data
Preconditions	Live retrieval is simulated as failing/timing out; local cached data exists.
Test Steps	1. Trigger a data request while the backend is simulated as failing. 2. Observe fallback behavior.
Test Data	Simulated backend timeout with cache present
Expected Result	System detects and returns the cached data along with a stale-data flag.
Priority	Medium
Status	Not Executed

Test Case ID	TC-17-03
Use Case	UC-17: Retrieve Data
Test Scenario	Retrieve Data — E1: Live retrieval fails with no cached data available
Preconditions	Live retrieval is simulated as failing and no cached data exists for the requested item.
Test Steps	1. Trigger a data request while the backend is simulated as failing and no cache exists.
Test Data	Simulated backend failure, no cache
Expected Result	System reports the failure to the calling use case.
Priority	Medium
Status	Not Executed

Manage Events

UC-18: Browse Events

Test Case ID	TC-18-01
Use Case	UC-18: Browse Events
Test Scenario	Browse Events — Main Flow (Positive Scenario): Student browses approved campus events successfully
Preconditions	Student is authenticated.
Test Steps	1. Select the Events module. 2. Allow system to retrieve approved campus events (UC-17). 3. Browse listings, apply category filters, or view event details.
Test Data	N/A
Expected Result	Student successfully browses approved campus events, with filters and details available.
Priority	High
Status	Not Executed

Test Case ID	TC-18-02
Use Case	UC-18: Browse Events
Test Scenario	Browse Events — A1: Student is redirected to an externally hosted event's registration page
Preconditions	An event in the listing is hosted on an external platform.
Test Steps	1. Browse event listings. 2. Select an event hosted externally. 3. Complete the external registration/browser flow and return to NotifiED.
Test Data	Event flagged as externally hosted
Expected Result	System redirects the student to the external registration URL/browser tab; upon return, the student is back at the event listing.

Priority	Medium
Status	Not Executed

Test Case ID	TC-18-03
Use Case	UC-18: Browse Events
Test Scenario	Browse Events — A2: No approved events are currently available
Preconditions	Backend is simulated to return an empty event list.
Test Steps	1. Select the Events module when no approved events exist.
Test Data	Empty event dataset
Expected Result	System displays 'No active events available.' Use case ends in success.
Priority	Low
Status	Not Executed

Test Case ID	TC-18-04
Use Case	UC-18: Browse Events
Test Scenario	Browse Events — E1: Event data fails to load due to a network error
Preconditions	Network request to load events is simulated as failing.
Test Steps	1. Select the Events module while the network request is simulated as failing.
Test Data	Simulated network failure
Expected Result	System displays an event loading error with a Retry button.
Priority	Medium
Status	Not Executed

UC-19: Register for Event

Test Case ID	TC-19-01
Use Case	UC-19: Register for Event
Test Scenario	Register for Event — Main Flow (Positive Scenario): Student successfully registers for an open event
Preconditions	Student is authenticated; the selected event exists and is open for registration; student has a completed profile.
Test Steps	1. From Event Details, select Register for Event. 2. Allow system to process enrollment (UC-20). 3. Review the resulting registration status (UC-22).

Test Data	Open event; completed student profile
Expected Result	Registration record is created/confirmed in the backend, and the confirmed status is displayed to the student.
Priority	High
Status	Not Executed

Test Case ID	TC-19-02
Use Case	UC-19: Register for Event
Test Scenario	Register for Event — A1: Student attempts to register for an event they are already registered for
Preconditions	Student has an existing registration record for the selected event.
Test Steps	1. Select Register for Event on an event the student already registered for.
Test Data	Event with a pre-existing registration for this student
Expected Result	System updates the UI to display 'Already Registered / View Ticket.' Use case ends in success.
Priority	Medium
Status	Not Executed

Test Case ID	TC-19-03
Use Case	UC-19: Register for Event
Test Scenario	Register for Event — E1: Registration fails because the event is full or registration is closed
Preconditions	Selected event has reached capacity or its registration period has ended.
Test Steps	1. Select Register for Event on a full/closed event.
Test Data	Event at capacity or past registration deadline
Expected Result	System notifies the student that registration is closed; no registration record is created.
Priority	High
Status	Not Executed

Test Case ID	TC-19-04
Use Case	UC-19: Register for Event
Test Scenario	Register for Event — E2: Registration submission fails due to a network error
Preconditions	Student submits a registration while the network request is simulated as timing out.

Test Steps	1. Select Register for Event. 2. Simulate a network timeout during submission.
Test Data	Simulated network timeout
Expected Result	System alerts the student that registration failed to submit; backend attendee state remains unchanged.
Priority	High
Status	Not Executed

UC-20: Confirm Registration

Test Case ID	TC-20-01
Use Case	UC-20: Confirm Registration
Test Scenario	Confirm Registration — Main Flow (Positive Scenario): Registration is confirmed and recorded successfully
Preconditions	Student has initiated a registration request for an open event.
Test Steps	1. Trigger the registration request (via UC-19). 2. System sends the request to the Notified Web Team. 3. Notified Web Team verifies capacity/eligibility and creates the record.
Test Data	Open event, eligible student
Expected Result	Notified Web Team returns a confirmation, and the system updates the UI to reflect the confirmed registration.
Priority	High
Status	Not Executed

Test Case ID	TC-20-02
Use Case	UC-20: Confirm Registration
Test Scenario	Confirm Registration — A1: Duplicate registration is detected and handled
Preconditions	Student already has a registration record for the selected event.
Test Steps	1. Trigger a registration request for an event already registered for.
Test Data	Pre-existing registration record
Expected Result	Notified Web Team detects the existing record; system updates the UI to display 'Already Registered.'
Priority	Medium
Status	Not Executed

Test Case ID	TC-20-03
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Use Case	UC-20: Confirm Registration
Test Scenario	Confirm Registration — E1: Registration fails because the event is full or closed
Preconditions	Event capacity is reached or the registration period has ended.
Test Steps	1. Trigger a registration request for a full/closed event.
Test Data	Event at capacity or closed
Expected Result	System notifies the student that registration is closed.
Priority	High
Status	Not Executed

Test Case ID	TC-20-04
Use Case	UC-20: Confirm Registration
Test Scenario	Confirm Registration — E2: Registration request times out
Preconditions	Request to the NotifiED Web Team is simulated as timing out.
Test Steps	1. Trigger a registration request. 2. Simulate a timeout to the NotifiED Web Team.
Test Data	Simulated backend timeout
Expected Result	System alerts the student that registration failed to submit.
Priority	High
Status	Not Executed

UC-21: Scan QR

Test Case ID	TC-21-01
Use Case	UC-21: Scan QR
Test Scenario	Scan QR — Main Flow (Positive Scenario): Student scans a valid event QR code and views event details
Preconditions	Student is authenticated; has app access; a valid NotifiED event QR code is available; camera permission is granted.
Test Steps	1. Select Scan QR. 2. Allow camera to activate and scan a valid event QR code. 3. Observe validation and event retrieval. 4. Review the displayed event details. 5. Select Register for Event if open.
Test Data	Valid event QR code
Expected Result	System validates the QR code, retrieves and displays the corresponding event details, and allows proceeding to registration (UC-19) if open.
Priority	High

Status	Not Executed
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Test Case ID	TC-21-02
Use Case	UC-21: Scan QR
Test Scenario	Scan QR — A1: Student scans an invalid or unsupported QR code
Preconditions	Camera and QR scanner are active.
Test Steps	1. Select Scan QR. 2. Scan a QR code that is invalid or unsupported.
Test Data	Invalid/unsupported QR code (e.g., random URL QR)
Expected Result	System displays an 'Invalid QR Code' message and prompts the student to scan another.
Priority	Medium
Status	Not Executed

Test Case ID	TC-21-03
Use Case	UC-21: Scan QR
Test Scenario	Scan QR — A2: Scanned QR code refers to an event that cannot be found
Preconditions	QR code is valid in format but the Notified Web Team cannot find a matching event.
Test Steps	1. Scan a validly formatted QR code with no matching event record.
Test Data	QR code referencing a deleted/non-existent event
Expected Result	System displays an 'Event Not Found' message. Use case ends in failure.
Priority	Medium
Status	Not Executed

Test Case ID	TC-21-04
Use Case	UC-21: Scan QR
Test Scenario	Scan QR — A3: Scanned event is no longer open for registration
Preconditions	Scanned QR code refers to a valid event that is closed for registration.
Test Steps	1. Scan a QR code for an event that is closed for registration. 2. Attempt to proceed to registration.
Test Data	Valid event, registration closed
Expected Result	System displays the event information and indicates registration is closed. Use case ends in success without proceeding to Register for Event.

Priority	Medium
Status	Not Executed

Test Case ID	TC-21-05
Use Case	UC-21: Scan QR
Test Scenario	Scan QR — E1: Student denies camera permission
Preconditions	Student selects Scan QR and is prompted for camera permission.
Test Steps	1. Select Scan QR. 2. Deny the camera permission request.
Test Data	Camera permission denied
Expected Result	System informs the student that camera permission is required to scan a QR code.
Priority	Medium
Status	Not Executed

Test Case ID	TC-21-06
Use Case	UC-21: Scan QR
Test Scenario	Scan QR — E2: Network error occurs while retrieving scanned event information
Preconditions	QR code is scanned and validated; network request to retrieve event data is simulated as failing.
Test Steps	1. Scan a valid QR code. 2. Simulate a network/server error during event info retrieval.
Test Data	Simulated network/server error
Expected Result	System displays an error message with an option to retry.
Priority	Medium
Status	Not Executed

UC-22: View Registration Status

Test Case ID	TC-22-01
Use Case	UC-22: View Registration Status
Test Scenario	View Registration Status — Main Flow (Positive Scenario): Student views their current registration status
Preconditions	Student is authenticated.
Test Steps	1. Select an event the student has registered for. 2. Allow system to request the registration record from the Notified Web Team.

Test Data	Event with existing registration record
Expected Result	NotifiED Web Team returns and system displays the current status (e.g., Confirmed, Waitlisted, Cancelled).
Priority	High
Status	Not Executed

Test Case ID	TC-22-02
Use Case	UC-22: View Registration Status
Test Scenario	View Registration Status — E1: Registration status fails to retrieve
Preconditions	Request to the NotifiED Web Team is simulated as failing.
Test Steps	1. Select a registered event. 2. Simulate a failure retrieving the registration record.
Test Data	Simulated backend failure
Expected Result	System displays a 'Status unavailable' message with a Retry option.
Priority	Medium
Status	Not Executed

AI Chatbot

UC-23: Ask Question

Test Case ID	TC-23-01
Use Case	UC-23: Ask Question
Test Scenario	Ask Question — Main Flow (Positive Scenario): Student receives an AI-generated answer to a campus-related question
Preconditions	Student is authenticated; AI feature is active.
Test Steps	1. Open the AI Chatbot. 2. Submit a campus-related natural language question. 3. Allow the system to route the request and retrieve context.
Test Data	e.g., 'When is the next enrollment period?'
Expected Result	AI Assistant processes the data and generates a natural-language answer, which is presented to the student.
Priority	High
Status	Not Executed

Test Case ID	TC-23-02
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Use Case	UC-23: Ask Question
Test Scenario	Ask Question — A1: Out-of-scope question is redirected toward campus topics
Preconditions	Student is authenticated; AI feature is active.
Test Steps	1. Submit a question unrelated to campus topics. 2. Observe the system's boundary handling. 3. Rephrase toward a campus-related topic.
Test Data	e.g., 'What's the weather like in Paris?'
Expected Result	System detects the query is out of scope; AI Assistant prompts the student to rephrase toward campus topics and resumes normally once a valid query is entered.
Priority	Medium
Status	Not Executed

Test Case ID	TC-23-03
Use Case	UC-23: Ask Question
Test Scenario	Ask Question — E1: AI cannot answer due to insufficient context
Preconditions	Student submits a campus-related question for which supporting context is unavailable.
Test Steps	1. Submit a question about data that is not present in the system. 2. Observe system response.
Test Data	Question referencing non-existent data
Expected Result	System displays a message stating that context is insufficient to form an answer.
Priority	Medium
Status	Not Executed

Test Case ID	TC-23-04
Use Case	UC-23: Ask Question
Test Scenario	Ask Question — E2: AI service is unavailable
Preconditions	Connection to the AI service is simulated as timing out or failing.
Test Steps	1. Submit a question while the AI service is simulated as down.
Test Data	Simulated AI service outage
Expected Result	System displays 'AI Assistant temporarily unavailable.'
Priority	Medium
Status	Not Executed

UC-24: Summarize Information

Test Case ID	TC-24-01
Use Case	UC-24: Summarize Information
Test Scenario	Summarize Information — Main Flow (Positive Scenario): Student receives an AI-generated summary of an item
Preconditions	Student is authenticated; AI feature is active; the item to summarize is available.
Test Steps	1. Select Summarize Information on an event, announcement, or email item. 2. Allow system to route content through the Notified Web Team for context.
Test Data	Existing event/announcement item
Expected Result	AI Assistant generates a concise summary, which is presented to the student.
Priority	High
Status	Not Executed

Test Case ID	TC-24-02
Use Case	UC-24: Summarize Information
Test Scenario	Summarize Information — E1: Summary cannot be generated due to insufficient context
Preconditions	Selected items have insufficient content/context for summarization.
Test Steps	1. Select Summarize Information on an item with minimal/no content.
Test Data	Item with empty or near-empty content
Expected Result	System displays a message stating that context is insufficient to summarize.
Priority	Medium
Status	Not Executed

Test Case ID	TC-24-03
Use Case	UC-24: Summarize Information
Test Scenario	Summarize Information — E2: AI service is unavailable during summarization
Preconditions	Connection to the AI service is simulated as timing out or failing.
Test Steps	1. Select Summarize Information while the AI service is simulated as down.
Test Data	Simulated AI service outage
Expected Result	System displays 'AI Assistant temporarily unavailable.
Priority	Medium
Status	Not Executed

